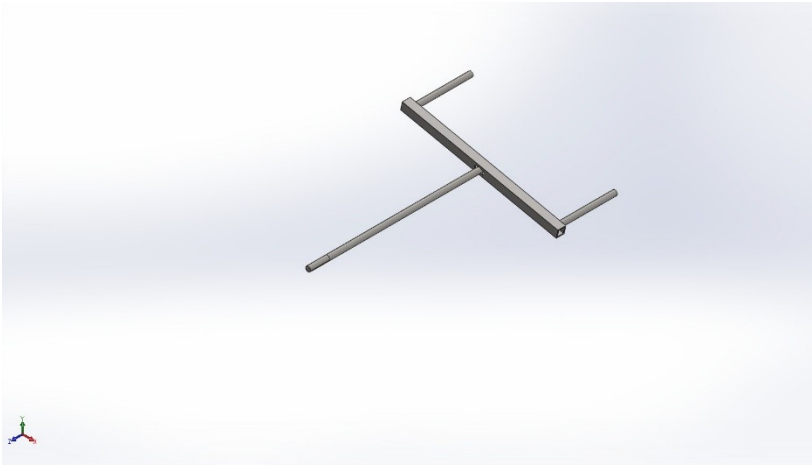




Description
No Data

Simulation of Tilty Loader

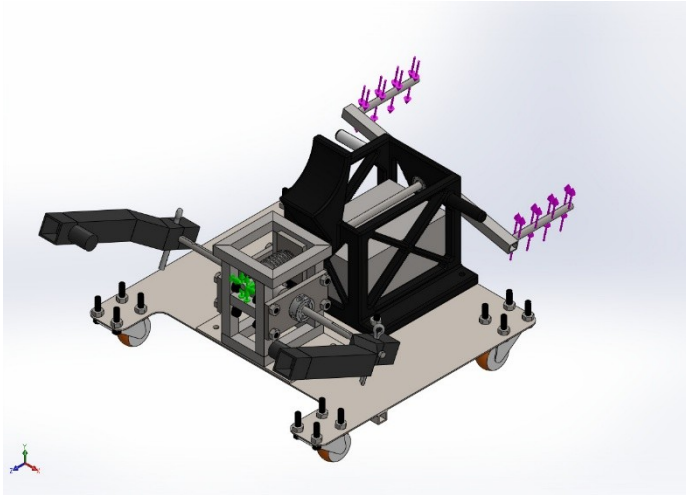
Date: Tuesday, December 9, 2025
Designer: Solidworks
Study name: Static 1
Analysis type: Static

Table of Contents

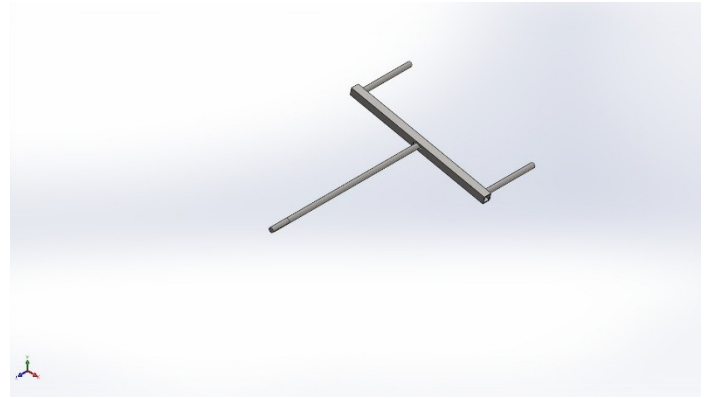
Description.....	1
Assumptions.....	2
Model Information.....	3
Study Properties.....	4
Units.....	4
Material Properties.....	5
Loads and Fixtures.....	6
Connector Definitions.....	7
Interaction Information.....	7
Mesh information.....	7
Sensor Details.....	7
Resultant Forces.....	8
Beams.....	8
Study Results.....	9
Conclusion.....	16



Assumptions



Original Model

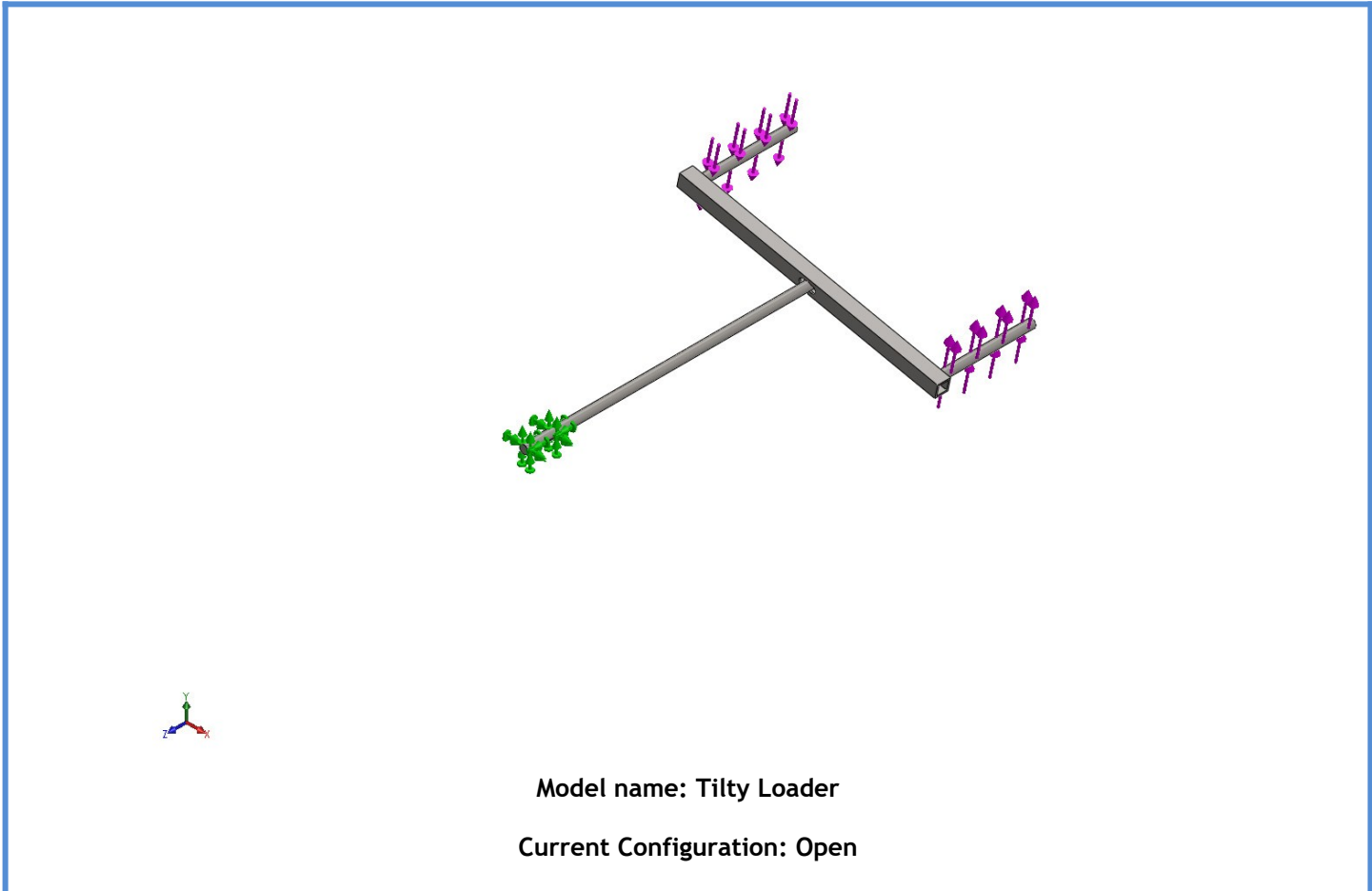


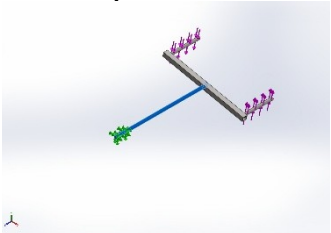
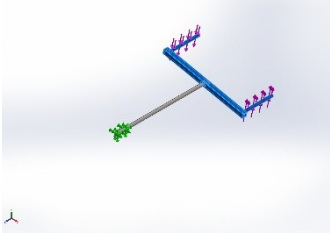
Model Analyzed





Model Information



Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line1 	Solid Body	Mass:0.112921 kg Volume:1.42938e-05 m^3 Density:7,900 kg/m^3 Weight:1.10663 N	G:\.shortcut-targets-by-id\1zACeN3N5D587UYBW02SNql2ed6qlvch\387 Team Project\CAD\Tilty Loader\HandleShaft.SLDPRT Dec 9 11:17:27 2025
Cut-Extrude1 	Solid Body	Mass:0.208788 kg Volume:2.64289e-05 m^3 Density:7,900 kg/m^3 Weight:2.04612 N	G:\.shortcut-targets-by-id\1zACeN3N5D587UYBW02SNql2ed6qlvch\387 Team Project\CAD\Tilty Loader\handleV2.SLDPRT Dec 8 13:12:49 2025



Study Properties

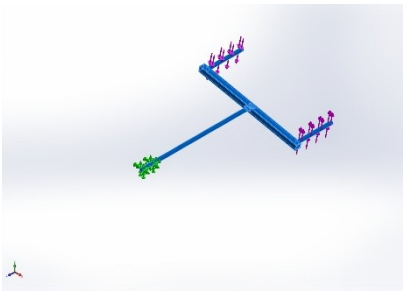
Study name	Static 1
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Contact penalty stiffness scale factor	1
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (G:\.shortcut-targets-by-id\1zACeN3N5D587UYBW02SNqI2ed6qllvch\387 Team Project\CAD\Tilty Loader)

Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

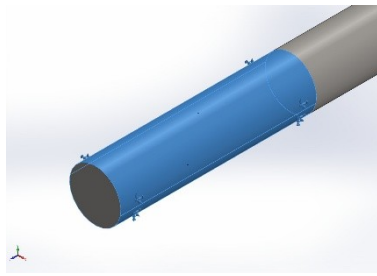


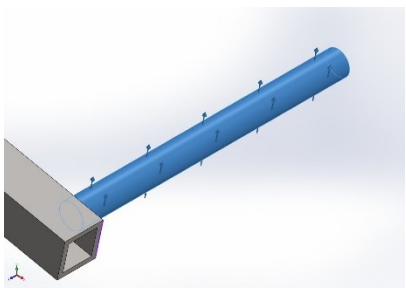
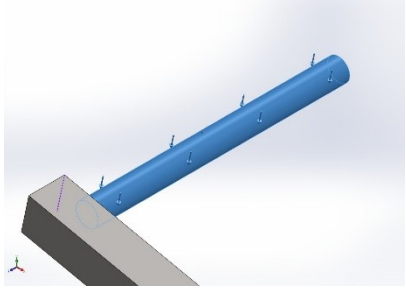
Material Properties

Model Reference	Properties	Components
	Name: AISI 1020 Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 3.51571e+08 N/m² Tensile strength: 4.20507e+08 N/m² Elastic modulus: 2e+11 N/m² Poisson's ratio: 0.29 Mass density: 7,900 kg/m³ Shear modulus: 7.7e+10 N/m² Thermal expansion coefficient: 1.5e-05 /Kelvin	SolidBody 1(Split Line1) (HandleShaft-1), SolidBody 1(Cut-Extrude1) (handleV2-1)
Curve Data:N/A		



Loads and Fixtures

Fixture name	Fixture Image	Fixture Details			
Fixed-1		Entities: 2 face(s) Type: Fixed Geometry			
Resultant Forces					
Components	X	Y	Z	Resultant	
Reaction force(N)	-8.25802e-05	3.47541e-05	-1.11312e-06	8.96024e-05	
Reaction Moment(N.m)	0	0	0	0	

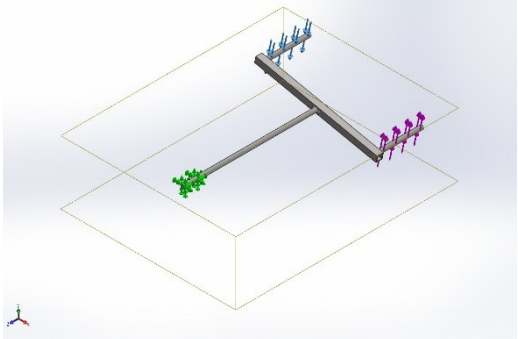
Load name	Load Image	Load Details		
Force-1		Entities:	1 face(s)	
		Reference:	Edge< 1 >	
		Type:	Apply force	
		Values:	---, ---, 7.5 lbf	
Force-2		Entities:	1 face(s)	
		Reference:	Edge< 1 >	
		Type:	Apply force	
		Values:	---, ---, -7.5 lbf	



Connector Definitions

No Data

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	0.092808 in
Minimum element size	0.0309357 in
Mesh Quality	High

Sensor Details

No Data



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-8.25802e-05	3.47541e-05	-1.11312e-06	8.96024e-05

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.000360979	8.03791e-05	-7.24543e-05	0.00037685

Free body moments

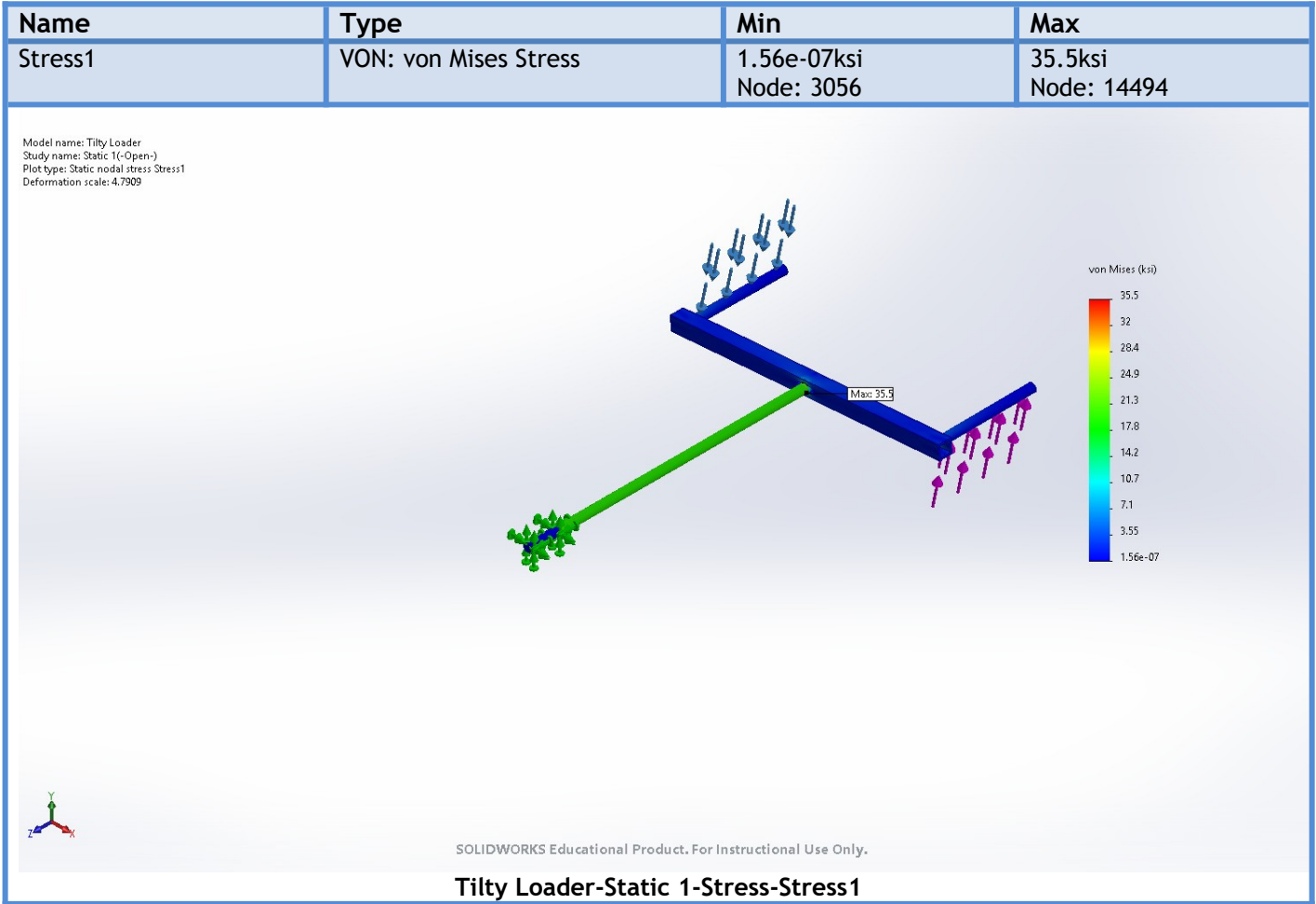
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

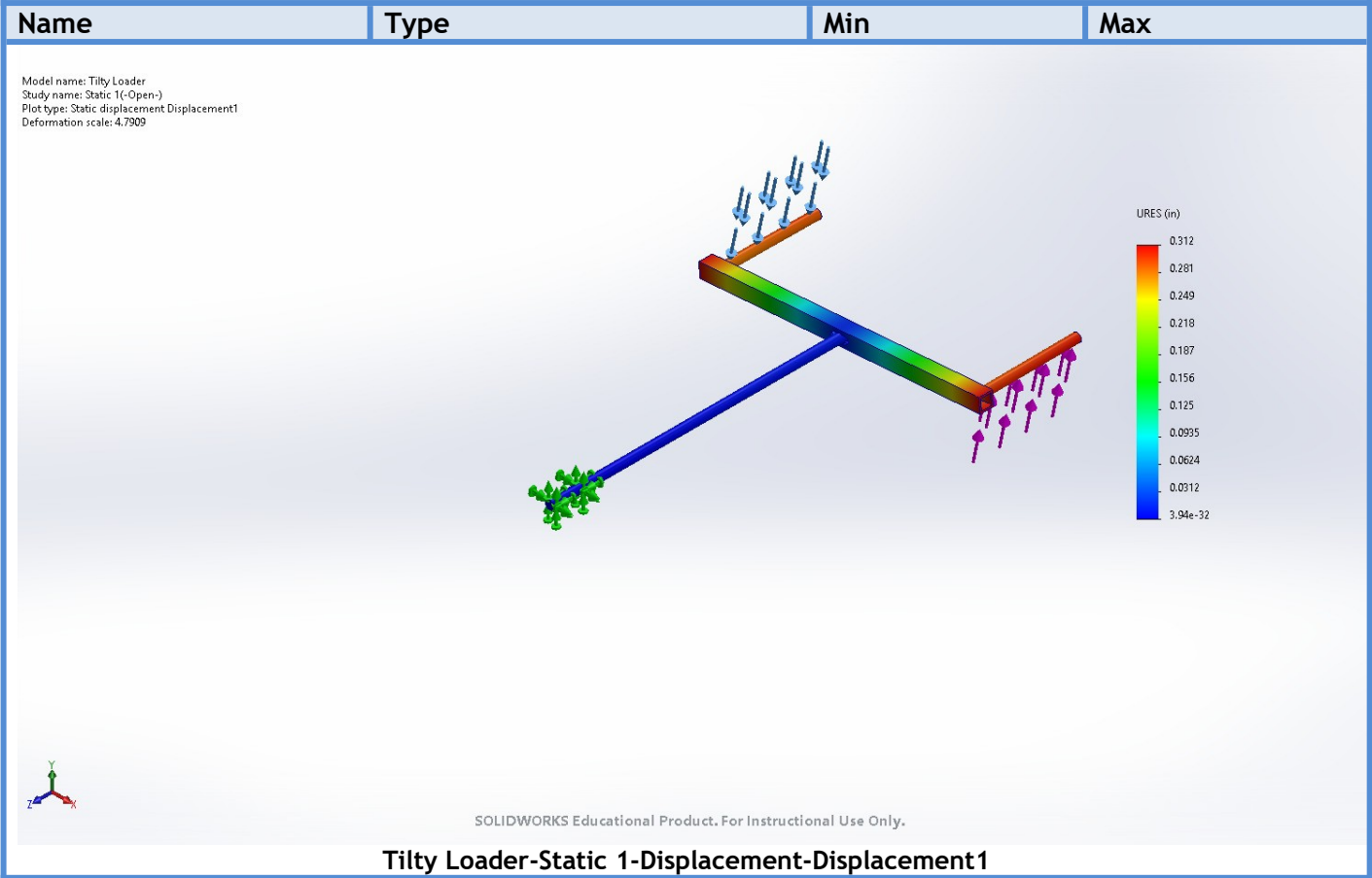


Study Results

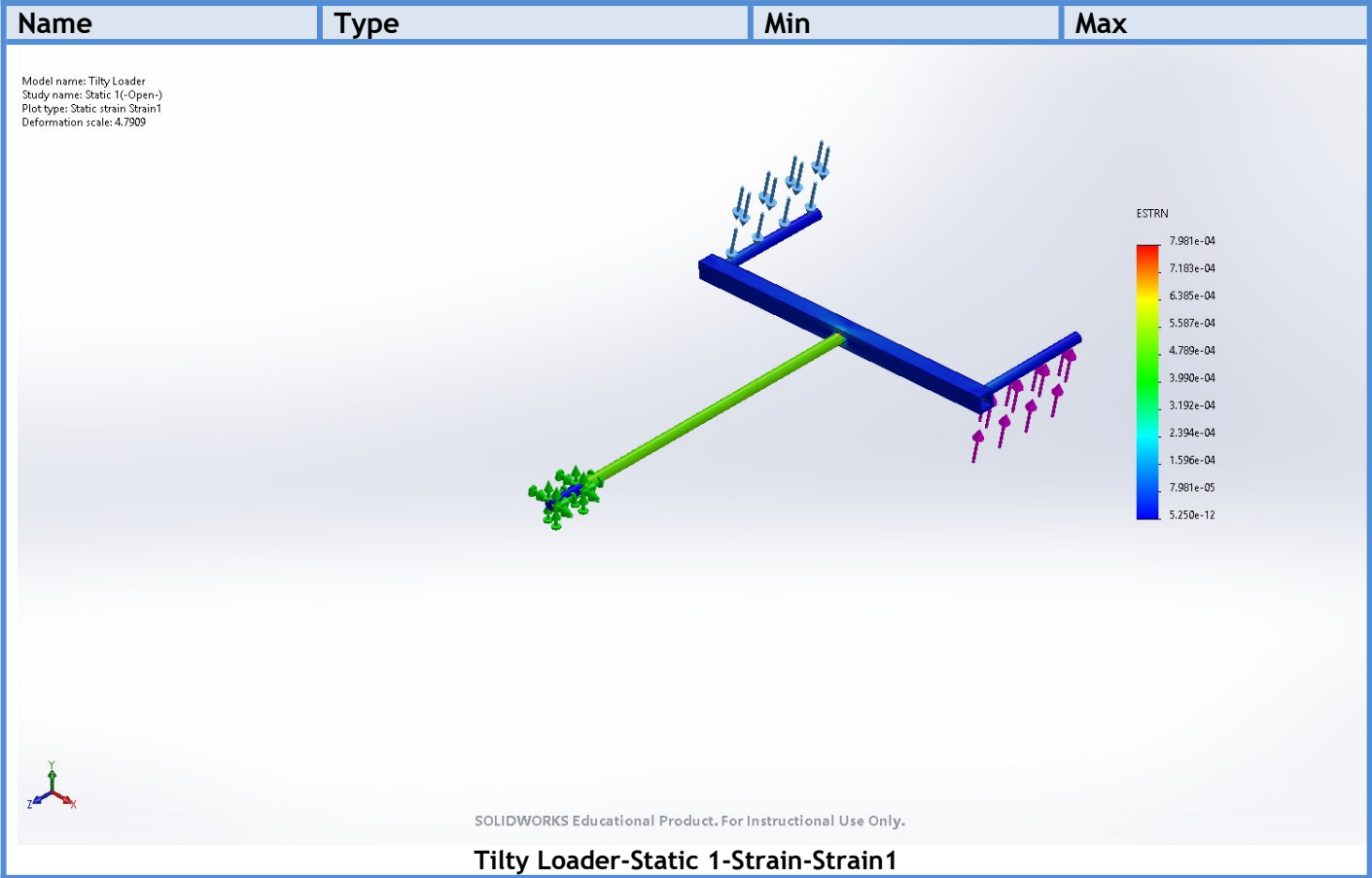


Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0in Node: 1	0.312in Node: 15292



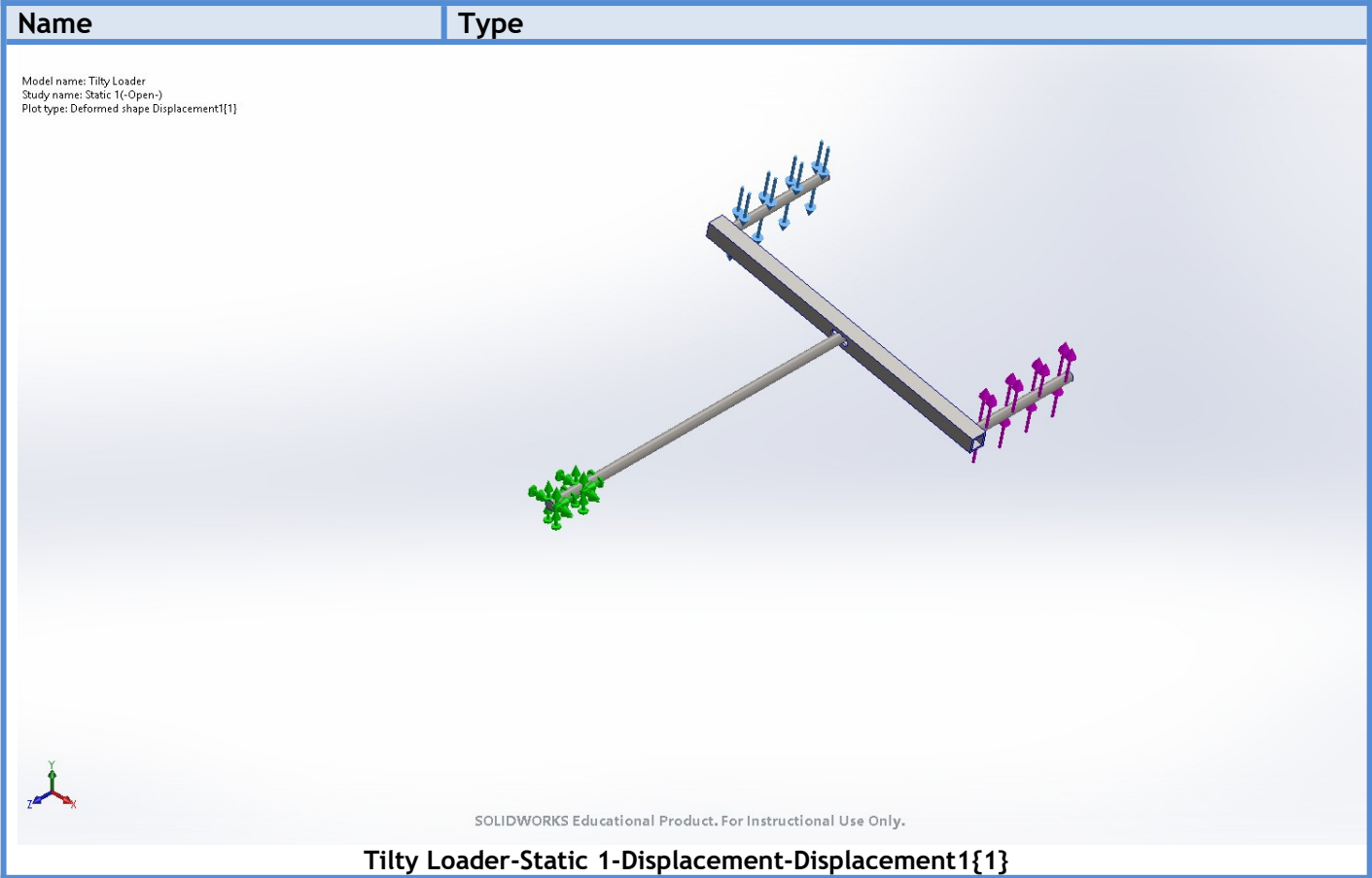


Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	5.250e-12 Element: 1966	7.981e-04 Element: 6078



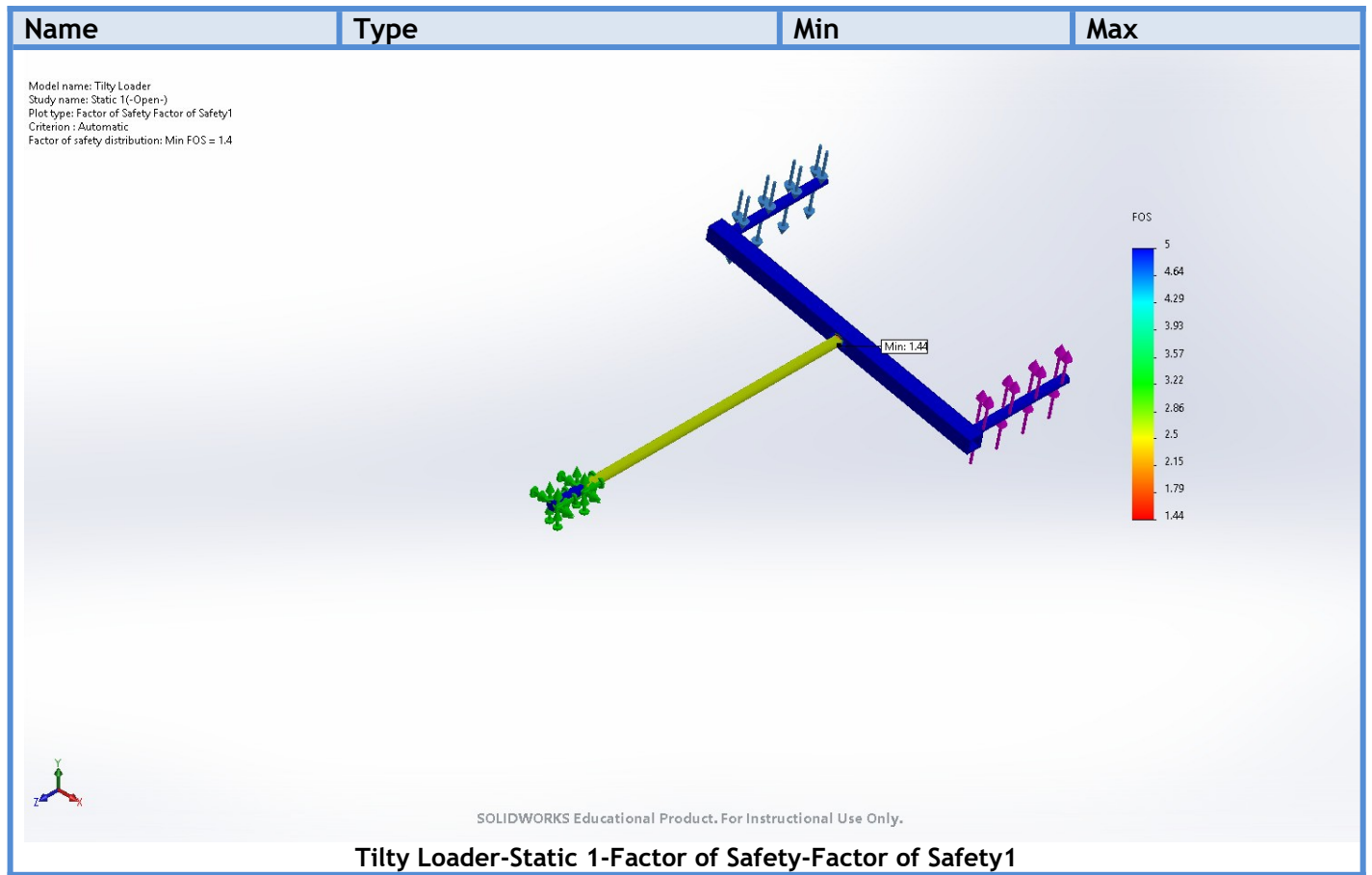
Name	Type
Displacement1{1}	Deformed shape





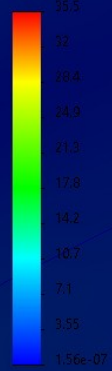
Name	Type	Min	Max
Factor of Safety1	Automatic	1.44 Node: 14494	5 Node: 2





Model name: Tilty Loader
Study name: Static 1(-Open-)
Plot type: Static nodal stress Stress1
Deformation scale: 4.7909

von Mises (ksi)



Max: 35.5

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Max Stress

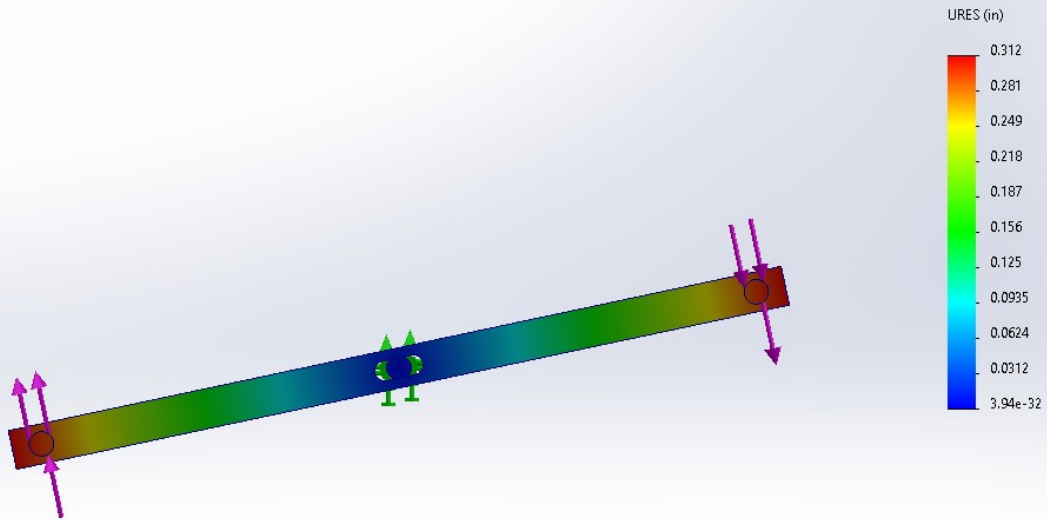


SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Tilty Loader

Model name: Tilty Loader
Study name: Static 1(-Open-)
Plot type: Static displacement Displacement1

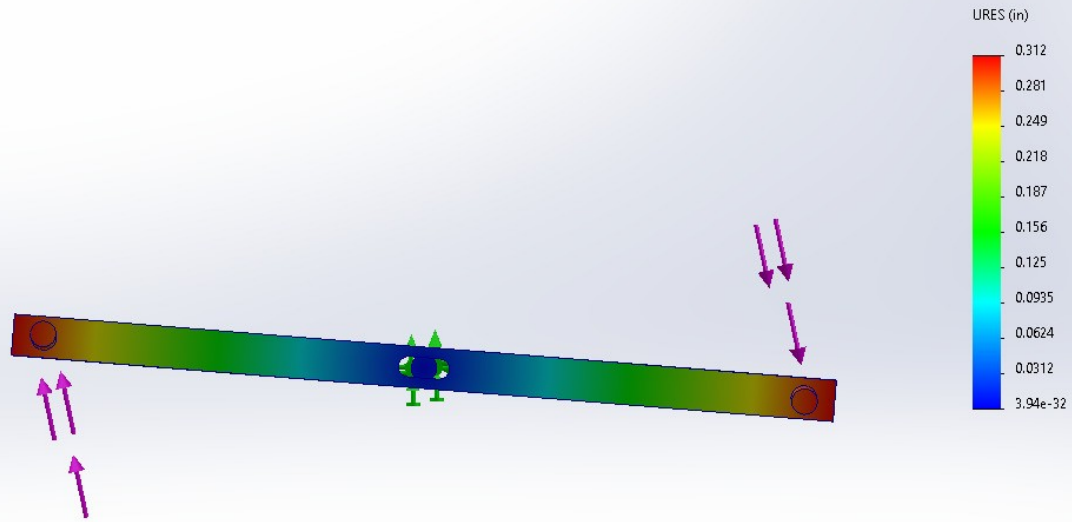


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Before Deformation



Model name: Tilty Loader
Study name: Static 1(-Open-)
Plot type: Static displacement Displacement1
Deformation scale: 4.7909



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After Deformation (Exaggerated)

Conclusion



SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Tilty Loader